

ReDataX

M3: Cross-Market Multiscale Model with Interaction

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1 Model role and testable hypothesis

M2 describes each market with a separate feature block, and its logit is additive in the input columns: the contribution of one market's state does not depend on the current state of another. M3 is built to test the following hypothesis.

Hypothesis 1.1. Joint market states contain information that cannot be represented by an additive function of the main M2 features. In other words, for target market A ,

$$\mathbb{P}(Y_{A,H} = 1 \mid X_A^{M2}, X^J) \neq \mathbb{P}(Y_{A,H} = 1 \mid X_A^{M2}),$$

where X^J is the vector of pairwise field products.

The M3-versus-M2 comparison is designed so that the J block is the only difference. The target market, forecast horizon, set of admissible observations, classification algorithm, temporal splits, and metric calculations remain unchanged. Any gain or deterioration can therefore be attributed to the added interactions.

2 Motivation: why M2 may not be enough

The M2 logit has the form

$$\eta_i^{M2} = \beta_0 + \beta_A^\top z_i^A + \beta_B^\top z_i^B,$$

where the contributions of the two markets are added independently. The model may increase its risk estimate when aggressive buying appears in the target market and also when it appears in a related market. What it cannot distinguish is a case in which two aligned flows reinforce risk beyond the sum of their isolated effects, or a case in which opposing flows dampen that risk.

Actual trade flow can be more intricate. An impulse in the target market may be dangerous only when a related market confirms it; two flows in the same direction may reinforce one another, while diverging flows may point to noise or a temporary dislocation. The interaction may also be selective: it may appear only at short scales, where synchronization matters, or only at long scales, where a common background has time to build up. M2 cannot represent these possibilities. M3 tests them through products of fields evaluated at matching scales.

The interaction idea is borrowed from a classic statistical-physics problem, the Ising model. Its Hamiltonian contains a sum of pairwise products,

$$H_{\text{interaction}} = \sum_{\langle i,j \rangle} J_{ij} s_i s_j,$$

where s_i and s_j are spins on neighboring lattice sites and J_{ij} is their coupling strength. If the spins are aligned ($s_i s_j > 0$), the pair's contribution to the energy depends on the sign of J_{ij} : one type of interaction lowers the total energy, while the other raises it. The useful point is that the effect of one spin is not constant - it depends on the orientation of its neighbor.

ReDataX uses this expression as a feature-design principle. The financial market is not being identified with an equilibrium physical system; the analogy simply motivates a way of encoding state-dependent cross-market effects.

3 The M2 base state

M3 is layered on top of M2 without changing the original M2 feature space. Its data, scale system, and target variable are identical to those of the final REAL-05 cross-market experiment. The three observed markets are

$$\mathcal{S} = \{\text{BTCUSDT}, \text{ETHUSDT}, \text{ETHBTC}\}.$$

The one-second flow field for market s is

$$\phi_s(t) = \begin{cases} \frac{V_s^+(t) - V_s^-(t)}{V_s^+(t) + V_s^-(t)}, & \text{if total volume is positive,} \\ 0, & \text{when no trades occur.} \end{cases}$$

At scale $B \in \mathcal{B} = \{1, 2, 4, 8, 16, 32, 64\}$ seconds, the smoothed field is the moving average

$$\bar{\phi}_s^{(B)}(t) = \frac{1}{B} \sum_{u=0}^{B-1} \phi_s(t-u).$$

For target market $A \in \{\text{BTCUSDT}, \text{ETHUSDT}\}$, the orientation $d_A(t) = \text{sign } \phi_A(t)$ defines the aggressor-aligned markout at horizon $H = 5$ seconds:

$$m_{A,H}(t) = 10^4 d_A(t) \frac{P_A(t+H) - P_A(t)}{P_A(t)}, \quad Y_{A,H}(t) = \mathbf{1}\{m_{A,H}(t) > 0\}.$$

The M2 base vector contains 87 features: three current log-volume values $\ell_s(t) = \log(1 + V_s^+ + V_s^-)$, plus four transformations of the smoothed field for every market and scale. These transformations are target-aligned flow $h_{s,B}^{(A)} = d_A \bar{\phi}_s^{(B)}$, amplitude $r_{s,B} = |\bar{\phi}_s^{(B)}|$, square $a_{s,B} = (\bar{\phi}_s^{(B)})^2$, and fourth power $b_{s,B} = (\bar{\phi}_s^{(B)})^4$.

4 Explicit cross-market interactions

Definition 4.1. For an unordered pair of distinct markets $s \neq r$ and a scale $B \in \mathcal{B}$, the *pairwise interaction feature* is

$$j_{sr,B}(t) = \bar{\phi}_s^{(B)}(t) \bar{\phi}_r^{(B)}(t).$$

A positive value corresponds to flows pointing in the same direction, while a negative value corresponds to opposing flows. Its absolute value grows when both imbalances are strong.

The implementation uses three unordered pairs:

$$\mathcal{P} = \{(\text{BTCUSDT}, \text{ETHUSDT}), (\text{BTCUSDT}, \text{ETHBTC}), (\text{ETHUSDT}, \text{ETHBTC})\}.$$

For each pair, fields are multiplied only at matching scales; the full 7×7 matrix of cross-scale products is not constructed. The number of new features is

$$\dim X^J = |\mathcal{P}| \cdot |\mathcal{B}| = 3 \cdot 7 = 21,$$

and the total model dimensionality is

$$\dim X^{\text{M3}} = 87 + 21 = 108.$$

In `build_feature_matrices()`, the interaction is calculated from fields that have first been multiplied by the target-flow sign. Since $d_A^2 = 1$ for every admissible observation, the final feature does not depend on which market is selected as the target:

$$[d_A \bar{\phi}_s^{(B)}] [d_A \bar{\phi}_r^{(B)}] = \bar{\phi}_s^{(B)} \bar{\phi}_r^{(B)}.$$

Directional asymmetry - BTCUSDT $\leftarrow$ ETHUSDT versus ETHUSDT $\leftarrow$ BTCUSDT - arises only through different labels and, as a result, different estimated coefficients.

5 Statistical model and training

M3 retains all 87 M2 features and adds 21 interactions. As in the preceding models, logistic regression parameterizes the probability with a sigmoid:

$$\hat{p}_A(t) = \sigma(\eta_A^{\text{M3}}(t)), \quad \eta_A^{\text{M3}}(t) = \beta_0 + \beta^\top z_A^{\text{M2}}(t) + \sum_{(s,r) \in \mathcal{P}} \sum_{B \in \mathcal{B}} \gamma_{sr,B} z_{sr,B}(t),$$

where $z_{sr,B}$ is the standardized interaction $j_{sr,B}$, and $\gamma_{sr,B}$ is its coefficient, denoted $J[s, r, B]$ in the code. Model parameters are estimated by minimizing

$$\mathcal{L}(\beta, \gamma) = - \sum_{i \in \text{train}} [Y_i \log p_i + (1 - Y_i) \log(1 - p_i)] + \lambda(\|\beta\|_2^2 + \|\gamma\|_2^2).$$

The implementation uses `SGDClassifier` with logistic loss, L_2 regularization, averaged coefficients, and a separate `StandardScaler` for each classifier. The regularization parameter α is selected by mean daily AP on the development period from $\{10^{-5}, 10^{-4}, 10^{-3}\}$. M3, like M2, selects $\alpha = 10^{-3}$. Once the hyperparameter is fixed, the model is refitted on the combined training and development samples.

6 Experimental protocol

REAL-05 uses the following temporal split:

Training:	06.01.2025–19.01.2025,
Development:	20.01.2025–26.01.2025,
Final test:	27.01.2025–02.02.2025.

M2 and M3 receive exactly the same final-test rows. The primary metrics are ROC-AUC, Average Precision (AP), the Brier score, and positive-class lift in the top 10% of predictions. The main criterion is the daily AP difference between M3 and M2. Uncertainty is estimated by resampling the seven test days with replacement using 5,000 bootstrap replications. “Brier improvement” is defined as $\text{Brier}_{M2} - \text{Brier}_{M3}$.

Table 1: REAL-05B: mean daily M3-minus-M2 difference.

Market	Metric	Difference	95% bootstrap interval
BTCUSDT	ROC-AUC	-0.000075	[-0.000191; 0.000018]
BTCUSDT	AP	0.000259	[0.000113; 0.000396]
BTCUSDT	Brier improvement	-0.000057	[-0.000173; 0.000056]
BTCUSDT	top-decile lift	0.000382	[-0.001617; 0.002497]
ETHUSDT	ROC-AUC	-0.000284	[-0.000471; -0.000096]
ETHUSDT	AP	-0.000229	[-0.000428; -0.000032]
ETHUSDT	Brier improvement	-0.000132	[-0.000196; -0.000077]
ETHUSDT	top-decile lift	-0.000257	[-0.001045; 0.000810]

7 Results

Table 1 reports the mean daily M3-minus-M2 differences over the REAL-05B final test.

For BTCUSDT, mean daily AP increases by 0.000259 and the difference is positive on six of seven days. ROC-AUC and the Brier score do not improve, however, and the top-decile lift interval includes zero. The AP effect is roughly two orders of magnitude smaller than the M2-over-M1 improvement. For ETHUSDT, M3 worsens ROC-AUC, AP, and the Brier score; its AP difference is positive on only one test day.

Result 7.1. The general added value of explicit pairwise interactions is not supported. The small market-specific effect for BTCUSDT does not make up for the lack of agreement across metrics and the deterioration on ETHUSDT. M3 is rejected as the common final model, while M2 without J remains the selected specification.

8 Limitations and reproducibility

M3 inherits all limitations of M0–M2 and adds several of its own. Observed flow alignment does not prove that one market influences another: a common external factor may move both flows at once and create the appearance of interaction. The interaction form is fixed in advance as a bilinear product at matching scales, so it does not cover other nonlinearities or relationships between different scales.

The coefficients $\gamma_{st,B}$ are correlated because the windows are nested and the model uses L_2 regularization, ruling out independent interpretation. The seven-day final period limits the extent to which the result can be transferred to other market regimes. Only equal-scale pairs are tested. The `aggTrades` data contain neither the order book nor platform inventory or actual execution costs. The result concerns observed flow and five-second markout; longer horizons require a separate study.

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